Post-Quantum Attestation and Intrusion-Aware CAN-FD Networking with Selective Recovery and Edge Anomaly Detection

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Abstract—This project studies post-quantum ECU attestation, reliable transport of a large protected object and layered intrusion detection for CAN-FD networks. A 2,420-byte ML-DSA-44 signature and 16-byte Ascon tag form a 2,436-byte protected object, fragmented into 41 payloads with a 4-byte transport header and 60-byte data region. A missing-frame bitmap supports selective retransmission over at most three recovery rounds. Runtime checks combine authenticated framing, replay control, discrete invariants and a 14-dimensional LSTM autoencoder. The study reviews existing code and archived artifacts rather than generating new measurements. In the manifest-anchored software-laboratory archive, all twenty canonical omission trials report successful recovery, with scenario means of 20.878–33.275 ms; selected negative cases exceed the experimental 50 ms budget. Saved IDS evaluations show that high ROC-AUC can coexist with zero recall or excessive false-positive rates at deployed thresholds. Actual committed ONNX files occupy 108,879 bytes in FP32 and 39,254 bytes in dynamic INT8. Conflicting metadata, deterministic benchmark keys and software timing limit broader claims. The contribution is an inspectable architecture and qualified evaluation, with secure lifecycle integration, calibrated detection and physical CAN-FD validation remaining future work.

Index Terms—CAN-FD, ML-DSA-44, Ascon-AEAD128, selective retransmission, intrusion detection, federated learning.

I. INTRODUCTION

In-vehicle networks carry measurements, commands and diagnostics among electronic control units (ECUs). The project considers four logical domains—powertrain, chassis, body and comfort, and infotainment—with a central in-vehicle security gateway (CISG). CAN identifiers support arbitration and application routing; they do not by themselves establish the identity or integrity of a sender. A security design must therefore address both unauthorized injection and apparently valid messages produced by a compromised enrolled node.

The project combines post-quantum boot attestation, authenticated runtime traffic, bounded selective retransmission and an intrusion detection system (IDS). These mechanisms address different failure modes. A lattice signature authenticates an attestation transcript, authenticated encryption detects message modification, recovery handles omitted fragments, and physical consistency checks seek attacks that remain cryptographically valid.

This study documents an implemented research prototype and its archived evidence. The principal result is a recoverable attestation transport and a layered detection design. The repository does not establish production readiness, vehicle-level functional safety certification, or timing guarantees on automotive hardware. Those distinctions determine how the results are interpreted.

II. RELATED WORK AND THREAT MODEL

ML-DSA is standardized in NIST FIPS 204 (2024). ML-DSA-44 uses a 2,420-byte signature, making direct placement in a single CAN-FD frame impossible. The project uses signatures at enrollment or admission rather than attaching a lattice signature to every runtime message. Ascon-AEAD128 is standardized in NIST SP 800-232 (2025) and provides authenticated encryption for constrained-device designs. Neither standard certifies this project's protocol composition [1], [2].

HKDF separates extraction of pseudorandom key material from expansion into context-bound keys. A bootstrap design can use HKDF-SHA256 with enrollment material and session context, but security depends on entropy, separation and lifecycle management; calling HKDF alone does not create a fresh secret [3].

CAN response-time analysis must include arbitration, blocking, higher-priority interference and transmission costs. Davis et al. [6] show why simplified schedulability arguments can be unsound. This study therefore does not equate observed laboratory means with worst-case response times.

The Real ORNL Automotive Dynamometer (ROAD) dataset contains over 3.5 hours of one vehicle's CAN data. Its publication distinguishes real recorded attack activity from simulated masquerade constructions. Using ROAD supports realistic signal relationships, but it does not establish performance across manufacturers, vehicle platforms or every attack family [4].

Federated averaging aggregates locally trained model parameters in proportion to client sample counts. The project applies this idea to five heterogeneous partitions of ROAD ambient data. The partitions emulate client silos within a training experiment; they are not five independently instrumented vehicles. The distinction matters when interpreting fleet generalization or privacy claims [5].

Cryptographic authentication and anomaly detection are complementary. A compromised legitimate ECU can possess valid keys while transmitting physically implausible values. Conversely, a cryptographically invalid frame can be rejected without invoking a learned detector. Evaluating both cases together under a single accuracy figure would obscure their different assumptions and denominators.

The design distinguishes an unenrolled injector, a network fault source, and a compromised enrolled sender. The injector may replay observed frames, forge tags, spoof logical-domain fields or flood a segment. The fault source

deliberately omits selected fragments or recovery controls. The enrolled attacker motivates invariant and kinematic checks because authentication alone cannot establish truthfulness of sensor data.

The central gateway and enrollment authority are trusted in the architecture. The fault relay is a laboratory instrument and should not possess application keys. Management HTTP and experiment commands are orchestration channels, not evidence of a CAN security protocol. Isolation of that management plane is an implementation requirement, not a property of CAN identifiers.

The current C++ benchmark source contains deterministic challenge generation and fixed test-key use. These choices constrain security claims for that execution path. The conceptual enrollment and fresh-key architecture must not be presented as fully realized by every revision of the benchmark. Physical key extraction, adversarial gateway compromise, electromagnetic fault physics and secure boot hardware are outside the demonstrated experiment.

III. ARCHITECTURE AND PROTOCOL

The CISG terminates each logical domain independently. In the intended path, an ECU responds to a boot challenge with a signed transcript protected under a bootstrap key. After complete reassembly, the gateway verifies the authenticated-encryption tag and signature before operational admission. Runtime messages then pass wire integrity checks, semantic invariants and an anomaly detector where appropriate.

The Docker laboratory separates ECU, gateway, relay and controller roles into services. Linux SocketCAN [8] and vxcan provide a software networking path for CAN frames. A vxcan link does not reproduce transceiver voltage, arbitration on a physical wire, oscillator drift, electromagnetic interference or a hardware receive timestamp. Deliberate omissions emulate transport loss; their names do not prove the physics of an inductive relay or alternator.

A. Attestation and key lifecycle

The architectural sequence starts with enrollment of a signature identity and bootstrap provisioning material. A challenge and session identifier bind an attestation response to a boot attempt. The ECU signs a canonical transcript containing its measured software identity and session context. The gateway should compare that measurement with an authorized enrollment value, not silently replace the expected value after a failure.

The protected attestation object in the frame design is the 2,420-byte signature ciphertext plus a 16-byte Ascon tag. The software measurement and transcript context are reconstructed or supplied separately for verification; this 2,436-byte length should not be described as a complete transmitted firmware image or a general certificate chain.

After verification, the intended protocol wraps a fresh operational key under the bootstrap key. A secure implementation must enforce nonce uniqueness per key and prevent replay across boot sessions. ML-DSA signing is not a key exchange mechanism and does not itself provide forward secrecy. The fixed-key and deterministic-nonce benchmark path requires correction and dedicated lifecycle validation before deployment claims are appropriate.

B. Runtime state commitment

The Phase 1 protected runtime frame is 64 bytes: an 8-byte clear header, 40 bytes of encrypted telemetry and a 16-byte authentication tag. The header carries version/type, sender ECU, domain, flags and a 32-bit big-endian counter. Authentication binds relevant header context to the ciphertext. The receiver should reject an invalid header or stale counter before committing replay state.

The 40-byte payload contains seven float32 values, two uint16 wheel speeds, an int16 temperature, a uint16 gear field and a uint32 diagnostic field. That representation is a project-specific telemetry mapping, not an OEM database definition. Recorded ROAD signals must be mapped and scaled explicitly before model inference.

A counter observation must not become trusted merely because it is large. Otherwise a forged high-counter frame could prevent acceptance of later legitimate traffic. Similarly, an unauthenticated sample must not update the learned detector's input history. The architecture separates parsing, authentication, replay-state commit and accepted-sample commit so these obligations can be evaluated independently.

The four-domain and attack-routing design views in Figs. 1 and 2 are reproduced from the supplied project figure collection. They express the intended architecture; they do not certify the current execution engine or establish empirical detection rates.

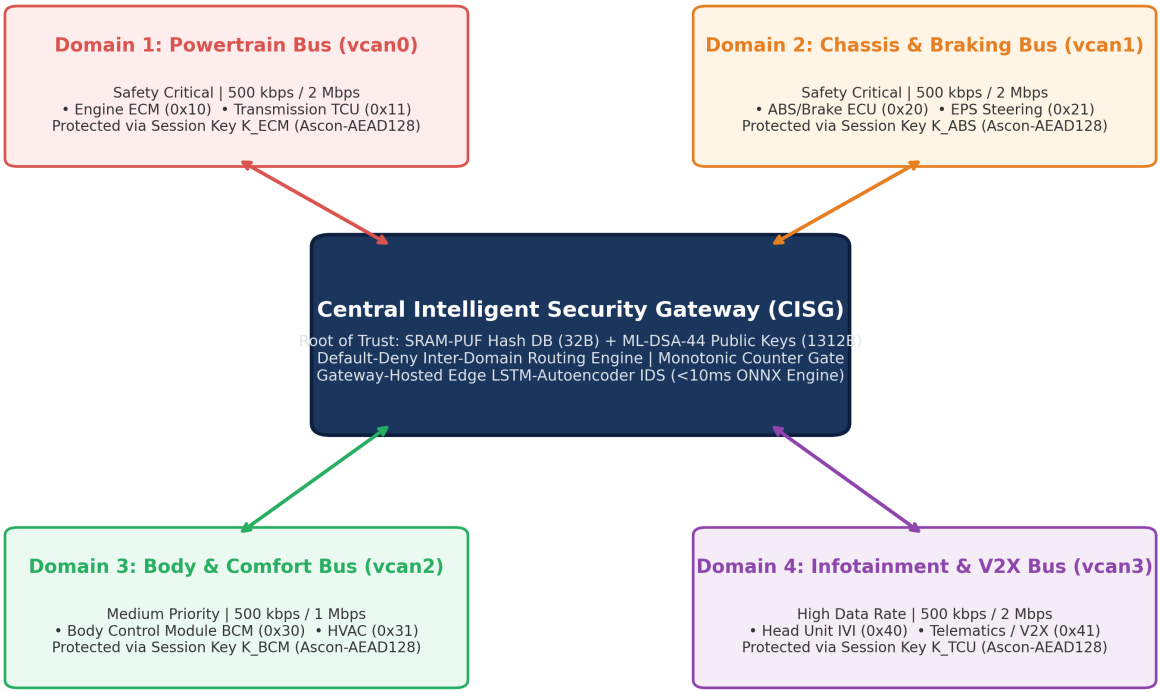


Fig. 1. Four-domain CISG architecture from the supplied figure document. The physical segmentation is a design view; laboratory transport uses virtual software interfaces.

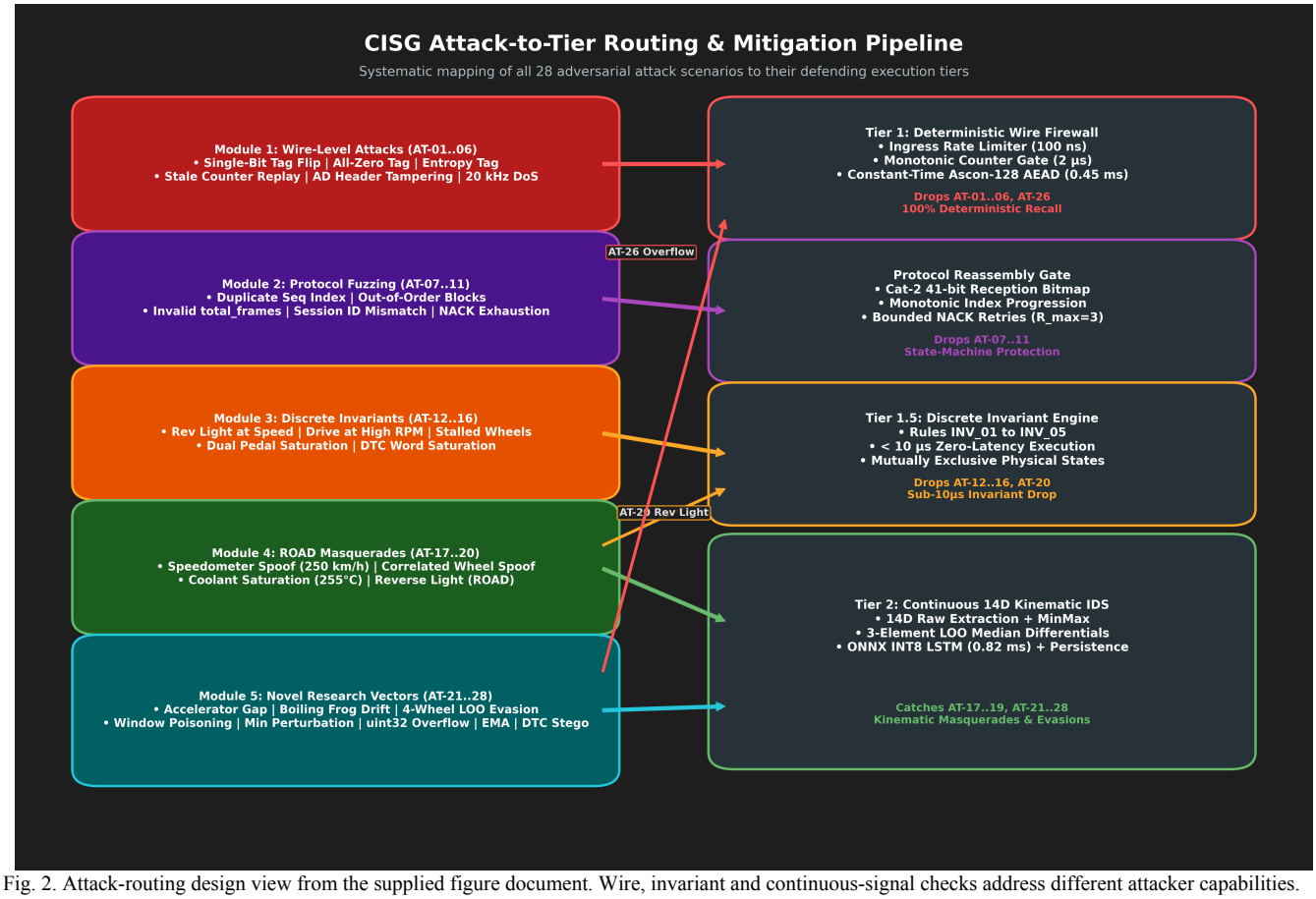


Fig. 2. Attack-routing design view from the supplied figure document. Wire, invariant and continuous-signal checks address different attacker capabilities. Embedded timing and recall annotations are historical design labels, not validated results; the numeric results are evaluated separately.

IV. SELECTIVE ATTESTATION RECOVERY

A. Indexed object fragmentation

Let P be the protected attestation length and q the fragment payload capacity. The frame count is N = ceil(P/q). With P = 2,436 and q = 60, N = 41. Frames 0 through 39 each contain 60 object bytes. Frame 40 contains the final 36 object bytes followed by 24 padding bytes. Each frame has a 4-byte transport header and a 64-byte CAN-FD payload.

The reference IndexedAttestationFrame header is session ID, frame index, total frame count and control flags, each one byte. Some supplied schematic pages show an earlier interpretation of those bytes; the implementation's header format >BBBB is the authority for this study. The one-byte transport session field is a shortened identifier and requires unambiguous association with full session context at the receiver.

The initial CAN data-field volume is 41 × 64 = 2,624 bytes. Protected-object efficiency is 2,436/2,624 = 92.835%. This ratio excludes arbitration, CRC, stuffing, acknowledgement and inter-frame spacing. Calling it physical-wire utilization or application sensor goodput would be incorrect.

B. Bitmap recovery

The receiver maintains a received-fragment set indexed from zero. A recovery bitmap sets bit i when fragment i is missing. For the relay-spike indices 14, 15 and 16, the mask is (1 << 14) | (1 << 15) | (1 << 16) = 0x000000000001c000. The NACK is carried on CAN ID 0x201. Hexadecimal serialization should retain the required 0x prefix and sixteen hex digits.

An ECU retransmits only requested fragments. The receiver verifies session association and bounds before placing data into the reassembly buffer. Duplicate fragments should not increase the count of unique recovered fragments. A late fragment from an older session must not satisfy a missing bit in the current object.

The experiment limits recovery to three rounds. A bounded retry policy provides a deterministic upper bound on the number of recovery attempts, but not a hard execution-time guarantee under arbitrary operating-system scheduling or bus interference. Persistent loss and missing recovery controls must terminate with distinct status codes so their causes remain visible.

C. Bounded termination and admission

The required statuses are SUCCESS, ERR_RETRY_EXHAUSTION, ERR_TIMEOUT_NACK_WAIT and ERR_CRYPTO_TAG_MISMATCH. The repository additionally records ERR_HEADER_OUT_OF_RANGE for malformed transport state. These are application logger labels used by the project; they are not universal status codes prescribed by ISO 26262.

The experiment compares recorded completion time with a 50,000 microsecond budget. A budget breach should inhibit operational admission according to the configured policy. That policy is an experimental design choice. ISO 26262 [7] defines a functional-safety framework and item-dependent timing concepts; it does not assign a universal 50 ms value to every ECU handshake.

The archived loss-exhaustion and NACK-timeout rows complete after roughly 94 and 91 ms respectively. Their error reporting demonstrates termination behavior, but does not prove transition to an actual vehicle safe state within 50 ms. A logger field such as FALLBACK_LOCKDOWN describes a software state label. Vehicle-level safety requires a hazard analysis, an allocated fault-handling budget, actuator behavior and hardware validation.

D. Timing model

A CAN-FD transmission-time estimate separates nominal-rate bits from data-rate bits: C = b_nom/R_nom + b_data/R_data. The model must specify frame format, stuffing assumptions, BRS behavior and inter-frame costs. Sleeping for a modeled frame duration on vxcan does not turn a virtual link into a physical CAN-FD controller.

For a complete transfer, latency includes admission setup, fragment transmission, receiver scheduling, recovery request delay, retransmission, cryptographic verification and policy evaluation. Measurements should identify their exact start and stop events. In the current C++ controller path, preparing the ECU before starting the gateway run can exclude work or admit early traffic; that timing boundary must be corrected before new end-to-end claims.

A formal response-time study additionally requires message priorities, release jitter, blocking and interference from the complete traffic set. The recurrence must use an appropriate CAN/CAN-FD scheduling model and demonstrate convergence or deadline failure. The archived analytical studies are useful exploration, but are not measurements of the deployed vehicle or a proof based solely on average utilization [6].

TABLE I
EVIDENCE SOURCES AND INTERPRETATION

Artifact	Evidence scope
31-row DockerLab CSV	Manifest hash matches archived file
35-row Phase 3/data CSV	Separate collection; not pooled
Baseline ONNX files	Current byte sizes inspected
IDS evaluation CSVs	Historical reported metrics
PMU probe	Unavailable; no counter claims

V. LAYERED INTRUSION DETECTION

Tier 1 rejects invalid framing, authentication failures, replay and excessive injection rates. Tier 1.5 evaluates discrete semantic invariants, such as consistency of a reverse-light state with the relevant operating state. Tier 2 evaluates continuous physical relationships using an LSTM autoencoder over a sequence window. This ordering reserves model inference for traffic that has passed lower-cost admissibility checks.

A sequence model can flag an unusual but benign operating condition as well as an attack. It therefore needs calibration on representative clean data and evaluation at a declared threshold. Receiver operating characteristic area under the curve (ROC-AUC) measures score ranking across thresholds; it does not establish good performance at the deployed threshold.

A. Feature calibration

The saved detector is described as a 14-dimensional physical-kinematics LSTM autoencoder. A window contains 30 steps. Feature extraction must use the saved scaler configuration, feature order and subsystem mapping. Mixing header/counter features with physical signals can reward detection of easy wire artifacts while obscuring the harder authenticated masquerade case.

The stored threshold configuration uses subsystem-specific P99.5 values: approximately 3.7043 for CAN ID 208, 5.0808 for 1760 and 2.7349 for 1255, with a global value of 3.6985. These thresholds are tied to that saved feature and residual pipeline. They cannot be reused after changing scaling, model weights, window alignment or channel semantics without recalibration.

The evaluation should preserve train, calibration and attack partitions independently. Temporal windows that overlap across partitions can leak information even when raw row indexes differ. An operational report should record the number of clean and attack windows, confusion counts, threshold and event-alignment rule. Saved aggregate metrics without those counts are secondary evidence and should not be promoted to a fleet-wide guarantee.

B. Trusted history and quarantine

A naive rolling buffer retains an anomalous sample after the attack has ended. With a 30-step window, one accepted bad sample may remain in as many as 29 subsequent windows. This explains the mechanism of post-attack smearing without implying that every model will actually produce 29 false alarms.

The proposed quarantine policy evaluates a candidate against trusted history and commits the candidate only after acceptance. A zero-order hold or equivalent state-preserving substitution can maintain a fixed input shape during rejection. The policy must define what happens during sustained rejection, initialization, legitimate rapid transients and missing data; otherwise holding old values can conceal real changes.

The saved comparative benchmark contains three ROAD scenarios. Post-attack smearing counts change from 29, 29 and 17 to zero, while total false-positive rates remain 1.18%, 2.20% and 0.00% in the quarantine mode. Thus zero post-attack smearing in these rows is not a zero false-positive guarantee. The CSV's ASIL_D_Status label is not accepted as evidence of certification.

C. Model deployment

The actual committed baseline ONNX artifacts are 108,879 bytes for FP32 and 39,254 bytes for dynamic INT8. This is a 63.95% reduction in file bytes for the inspected files. Historical onnx_export_metadata.json instead reports 50.44 KB and 72.86 KB; those values do not describe the current files and are not used as current artifact sizes.

The real_telemetry_onnx_evaluation.json artifact reports development-host means of 1.484 ms for eager PyTorch, 0.159 ms for ONNX FP32 and 0.198 ms for ONNX INT8. These figures describe a historical single-window evaluation. The same file explicitly calls its Raspberry Pi 4 tail-latency figures simulated. They cannot be described as measured target-board execution.

INT8 reduces the inspected file size but is slower than FP32 in that historical development evaluation. Quantization therefore has separate storage, numeric and runtime tradeoffs. One reported wheel-speed decision differs between INT8 and its reference among 5,000 evaluated windows; a broad claim of complete decision equivalence is unsupported.

D. Federated adaptation

The stored federated-learning metadata defines five clients, ten rounds, two local epochs per round, batch size 2,048 and learning rate 0.001. Each client partition contains 100,000 rows split into 80,000 training and 20,000 local test rows. The setup is a controlled multi-silo simulation using Flower NumPyClient and explicit server FedAvg.

For client k with n_k training samples and parameter vector w_k, weighted averaging forms w_next = sum_k (n_k / sum_j n_j) w_k. Uniform client sizes make these weights equal in the recorded setup. This averaging equation describes the aggregation operation; it does not guarantee convergence, privacy or resistance to poisoned client updates.

The metadata also separates round-validation captures, a 40,000-row benign calibration suite and a 30,000-row held-out reverse-domain partition. Those names document intended separation. A full reproduction must verify file hashes and window-level disjointness rather than relying solely on a metadata assertion.

The cross-silo matrix reports a held-out fleet MSE of 0.003948 for the global federated model. The five local-model values are 0.002710, 0.007358, 0.040995, 0.007360 and 0.007256. The global model improves on four locals but is worse than the highway local on this particular held-out partition. This supports a heterogeneous tradeoff rather than universal dominance.

The saved federated attack benchmark reports ROC-AUC values near one, yet filtered false-positive rates of 97.775% for speedometer spoofing and 99.539% for wheel-speed spoofing. At those operating thresholds the detector is unsuitable for direct operational blocking. AUC can remain high when threshold calibration is poor; the two observations are not contradictory.

The coolant row reports filtered recall 95.35%, filtered FPR 1.382% and filtered F1 0.9425. These results demonstrate why each scenario and operating point must be disclosed. The CSV's generic PASS label does not override its numeric metrics. Secure aggregation, poisoning defenses and production model-update governance are not established by the saved FedAvg experiment.

VI. EXPERIMENTAL METHOD

The current Dockerfile builds an Ubuntu 22.04 image with C++17 components, liboqs 0.12.0 and ONNX Runtime 1.20.1. The compose image is pqc-can-lab:cpp17. The repository also retains a Python orchestration and execution path, so the language and engine must be recorded for every experiment rather than inferred from the latest README.

The services include relay, gateway, domain ECUs, controller and observability components. The relay injects controlled omissions and the controller coordinates scenarios. The dashboard visualizes process events. Management messages, container logs and CAN receive

observations serve different purposes; dashboard animation alone is not a network capture or a cryptographic verification record.

The current compose file exposes management services and grants capabilities including NET_ADMIN and PERFMON to application services. Earlier claims of a minimal-capability or localhost-only management boundary do not describe every current service declaration. The report treats the setup as a research laboratory and records configuration hardening as future work.

The primary descriptive transport dataset is Phase 3/DockerLab/simulation_results/fault_recovery_runs.csv, containing 31 rows. Its SHA-256 is 49f1ec247961fa03d7cad f378e6c57e4fb19b4861c7177e79bd4f0e320a1e56b, matching the adjacent manifest entry. The manifest identifies a Python 3.10.12 experiment profile, Linux kernel 6.18.42-1-lts and a software SocketCAN/vxcan transport. This is archived evidence, not a new run of current C++ HEAD.

The separate Phase 3/data/fault_recovery_runs.csv has 35 rows and SHA-256 93dddec9f3fd0f08f82a816691d70ed05c 0e7ea207cd694d00c14646da41bb2a. It includes additional protocol rows and different timings. Those rows are not pooled with the manifest-anchored 31-row laboratory collection. The supplied old plots and performance narrative are likewise not substituted for the selected CSV.

The manifest reports PMU permission/unavailability and zero arrival rows while other files contain populated profiling or reconstructed latency records. Consequently, this study does not infer hardware counter measurements or a complete raw receive capture from the existence of those CSVs. Matching a run-CSV hash supports file association; it is not independent proof of every field's semantics.

The quantitative transport source is the archived manifest-associated CSV, not a rerun of current C++ HEAD [9]. Canonical statistics use rows 1–20 in five groups of four. The first four clean observations define the canonical baseline; later clean setup and corruption rows are not pooled into it. Sample standard deviations use the n−1 denominator. No confidence bound, hardware WCET or rare-failure rate is inferred from these small groups.

VII. RESULTS AND DISCUSSION

A. Canonical omissions

Table II summarizes the first twenty rows of the manifest-anchored laboratory CSV. Each of the five canonical scenarios has four repeated observations. All twenty rows report SUCCESS and no violation of the configured 50 ms budget. Means and sample standard deviations are recalculated directly from end_to_end_latency_us, converted to milliseconds.

The clean mean is 20.878 ms. Single-frame, late-frame, relay-spike and alternator-burst means are 30.432, 31.368, 31.913 and 33.275 ms respectively. Relay-spike observations have the largest dispersion among these groups, with a sample standard deviation of 7.100 ms. Four trials per group are insufficient to estimate rare-event tails or a worst-case execution bound.

The canonical scenario names denote deliberate fragment omissions. They do not establish a measured EMI waveform or a real alternator experiment. Successful reassembly under these omission patterns supports selective recovery behavior in the archived software laboratory. It does not establish safety integrity level, physical bus arbitration performance or timing portability to a microcontroller.

TABLE II
CANONICAL ARCHIVED TRANSPORT RESULTS

Scenario	n	Mean ms	Sample SD ms
Clean	4	20.878	0.921
Single drop	4	30.432	1.480
Late drop	4	31.368	1.575
Relay spike	4	31.913	7.100
Alternator burst	4	33.275	4.064

B. Failure cases

Rows 21–24 exercise persistent omission, missing NACK delivery, cryptographic corruption and malformed transport state. Their recorded terminal statuses are retry exhaustion, NACK timeout, tag mismatch and header out-of-range. The corresponding latencies are 93.989, 90.714, 30.076 and 60.376 ms. Three exceed the configured 50 ms budget.

Row 25 records a successful header-probe recovery at 36.721 ms. Rows 26–30 are labeled Monte Carlo Loss Sweep and report successful completion from 16.673 to 46.099 ms. Their repeated configuration and the timing-boundary concerns prevent treating them as an independently seeded reliability study. Row 31 is a separate clean setup observation at 28.889 ms and is excluded from the four-trial canonical clean group.

The correct conclusion is bounded functional recovery and differentiated failure reporting in the selected archive. It is not "all thirty-one runs meet the budget." Failed authentication and failure to finish within the budget remain visible even when a harness considers the expected negative test to have passed.

C. Wire attacks

The archived manifest records one replay, one forged-tag injection and one spoofed-domain injection. Their observed events are replay_rejected, runtime_tag_rejected and header_rejected respectively. The flood test records 300 transmitted frames and 202 rate_limit_rejected events. A 202/300 rejection fraction is 67.33%; it is not 100% rejection of every flooded frame.

All four manifest attack entries report zero other-domain receive observations. This is useful evidence for the tested topology and interval, but it does not prove that every current compose configuration enforces isolation or that all possible cross-domain attacks are blocked. The record should be read together with raw observer coverage and the selected network namespace topology.

The attack checks are deterministic probes with very small sample sizes. No ROC curve, vehicle-wide false-positive rate or statistical detection guarantee is derived from them. The high-level security property depends on authenticated state commit and fresh keys, which require separate review of the actual execution engine.

D. IDS operating points

The master_3tier_real_benchmark_leaderboard.csv records ONNX FP32 speedometer recall of 99.90% and FPR of 0.141%, but wheel-speed and coolant recall of 0.00% at the saved operating points. Their ROC-AUC values are 0.9961 and 0.9988. The numerical rows contradict any interpretation that all three attacks are reliably detected at those thresholds.

The invariant-gate rows vary too: reverse-light-off variants report recall of 100.00%, 84.94% and 85.70%. The architecture offers a useful separation of discrete and continuous checks, but aggregate "PASS" tags hide these differences. A research paper must report thresholded metrics alongside ranking metrics and state the evaluation source.

The standalone ONNX export evaluation reports near-perfect AUC on a selected 5,000-window-per-scenario test and one INT8 wheel-speed decision flip. The different evaluation files are not merged because they use different scopes and potentially different calibration pipelines. A future reproducible run should reconcile weights, scaler, threshold, labels and split hashes in one manifest.

E. Analytical studies

Phase 3/data/transport_protocol_benchmark_data.csv and provenance_metadata.json describe a separate analytical transport study. The metadata specifies seed 42, 10,000 trials per point, eleven loss points and three bitrate regimes. These model results explore protocol behavior under declared assumptions; they are not raw SocketCAN observations or measurements from CAN transceivers.

Protocol efficiency comparisons must hold the protected-object length, control accounting, retry policy and wire-cost model constant. For the proposed initial transfer, 92.835% is a data-field efficiency. A comparison with whole-object retransmission depends on the probability of object failure and the model of correlated loss. There is no context-free overhead reduction percentage valid for every channel.

The supplied analytical figures include platform scaling, schedulability and boot-storage predictions. They are excluded from the paper's empirical claims where target-board measurements and complete task-set assumptions are absent. Their appropriate role is to motivate a hardware validation plan and identify parameters that need measurement, rather than to certify deadlines.

TABLE III
REPORTED IDS OPERATING POINTS; BASELINE ONNX FP32 AND FILTERED FEDERATED RESULTS USE DISTINCT EVALUATIONS

Scenario	AUC	Recall	FPR
Baseline speed	0.9995	99.90%	0.141%
Baseline wheel	0.9961	0.00%	0.000%
Baseline coolant	0.9988	0.00%	0.000%
FL speed	0.9995	100%	97.775%
FL wheel	0.9999	100%	99.539%
FL coolant	0.9999	95.35%	1.382%

VIII. LIMITATIONS AND FUTURE WORK

The first limitation is provenance drift. The current checkout contains different result sets, source revisions and metadata values. This study anchors each quantitative statement to a named artifact and avoids treating the archived Python profile as a measurement of current C++ execution. No new experiments were run for the manuscript.

The second limitation is measurement scope. Kernel software timestamps, process clocks and modeled frame delays are not hardware timestamps or physical serialization measurements. The archived PMU probe reports unavailability. Claims about cycles, cache misses, zero major faults, fixed microsecond verification time or Cortex-A72 execution therefore require new raw measurements with a reproducible environment.

The third limitation is protocol realization. Fixed benchmark keys, deterministic challenge generation, possible nonce reuse, management exposure and an in-process fallback path prevent a blanket claim of production-secure orchestration. These limitations do not invalidate the frame arithmetic or the usefulness of the architecture, but they limit what the prototype demonstrates. A complete security argument must bind the selected engine to an audited key lifecycle.

A hardware campaign should use CAN-FD-capable controllers and transceivers, terminated physical domains, an independent bus analyzer and synchronized measurement boundaries. It should record arbitration interference, error frames, bus-off behavior and recovery timing under controlled load. Vehicle or dynamometer attack work requires an isolated bench and a defined safety procedure; the present archive is a software laboratory.

The cryptographic path should replace deterministic benchmark material with enrolled per-ECU secrets, context-separated derivation, unique nonces and durable anti-replay state. The C++ prepare/run race should be removed before measuring the entire handshake. The image digest, source hashes and raw observations should be generated together and stored as an immutable run bundle.

The IDS campaign should recalibrate thresholds using disjoint clean data, evaluate unseen operating conditions and report confusion counts and false alarms per unit time. Federated learning should be compared against local and centrally trained baselines at matched operating points, followed by poisoning and update-integrity studies. Any ISO 26262 claim would require an item definition, hazard analysis, allocated safety requirements and independent evidence beyond these experiments.

IX. CONCLUSION

The project demonstrates a technically coherent architecture for separating boot identity, reliable object transport and runtime anomaly detection. The 2,436-byte protected object fits into forty-one indexed CAN-FD payloads, and a compact missing-frame bitmap permits selective recovery. The archived canonical trial groups all report successful recovery, with mean completion times from 20.878 to 33.275 ms in the selected software laboratory.

The saved negative tests reveal budget breaches and distinct failure causes. The IDS evidence reveals a second important result: high ROC-AUC does not ensure useful recall or false-positive behavior at a chosen threshold. The inspected INT8 artifact is smaller than FP32, but historical development-host timings do not show an INT8 speed advantage. These findings identify concrete engineering choices for the next iteration.

The academic objectives are met at the level of an integrated design, inspectable software artifacts and a qualified analysis of existing observations. Hardware timing, secure production provisioning, cross-vehicle detection generalization and functional-safety assurance remain open. The most defensible next step is a unified reproducible run with corrected lifecycle and timing boundaries, followed by physical CAN-FD validation.

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